

DG32-2DOM - Preliminary Datasheet

The DG32-LITE motor-control SoC plus an INT8 attention accelerator on a second clock. Same lockstep core, boot path and motor peripheral set; adds a memory-mapped attention engine (0xE000_0000) that runs on `clk_fast_i` and reaches memory through clock-domain-crossing bridges. For firmware and hardware/PCB engineers.

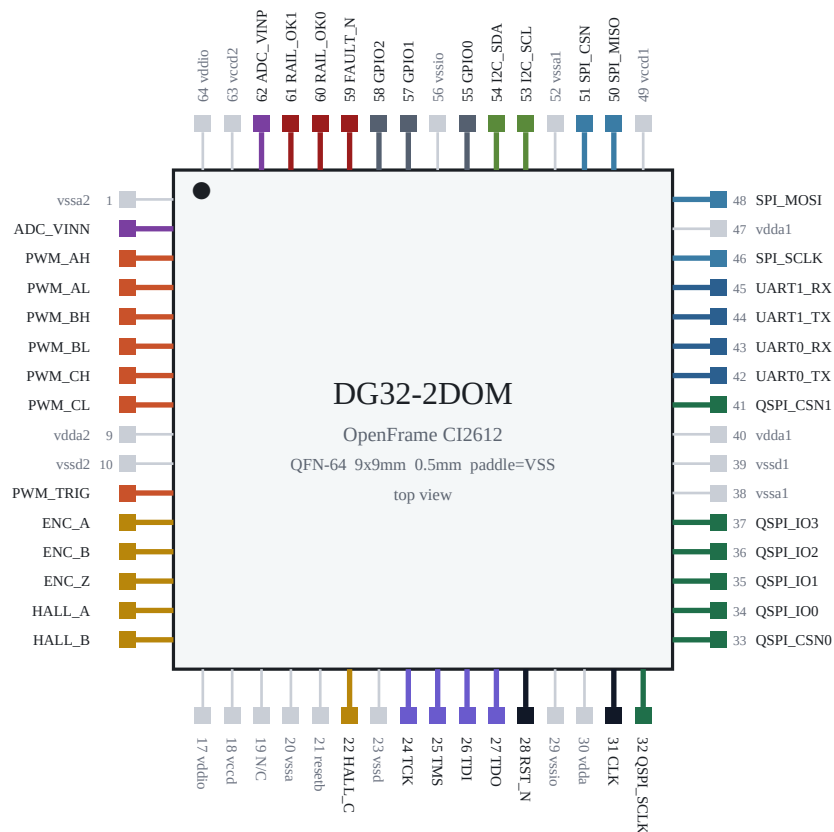
Part	Package	Process	Core	Clocks
DG32-2DOM	QFN-64, 9x9 mm	sky130A 130 nm	2x rv32imc + INT8 attn	50 / 114 MHz

Preliminary, pre-silicon. Functional and register detail is from the taped-out RTL and the hardened die. Electrical limits are design targets and sky130 process nominals pending first-silicon characterisation, and are marked where they are not yet measured. The flight-control core is frozen until silicon test.

1 Features

- Everything in DG32-LITE: dual rv32imc lockstep core, 32 KB SRAM, 64 KB boot ROM, QSPI-XIP, and the full motor peripheral set (PWM, DShot, encoder, CORDIC, SAR ADC) plus UART/SPI/I2C/GPIO.
- INT8 attention accelerator (0xE000_0000): one kick computes a band of query rows of one head end to end (QK^T , softmax/EXP, E.V, requantise to int8), bit-exact to the golden model.
- Attention datapath: shared LANES-wide MAC array, 256-entry u15 EXP table, 40-bit numerator (exact to $NK=131072$), one reciprocal per row; ~3242 cycles/row analytic at LANES=16.
- Second clock domain: the engine runs on `clk_fast_i` (114 MHz) and crosses to the 50 MHz CPU domain through 2-flop CDC bridges, so it never stalls the frozen control core.
- Own die (3400 x 4500 um) with the attention K/V buffers on sky130 SRAM macros next to the SAR analog corner.

2 Pinout - 64-pin QFN (top view)



The external pinout is the same 64-pin QFN as DG32-LITE: the attention engine is internal (it talks to memory over the on-chip bus and to the CPU over AXI-lite), so it adds no user pads. The only added clock, `clk_fast_i`, is a review item - either derived on-die from CLK or brought in on a spare pad for the CI2612 die.

3 Pin description

QFN pin 1 is at the upper-left; numbering runs counter-clockwise. User signals are the DG32 function on each OpenFrame user pad; power and ground pins are the harness pad-ring supplies. The exposed paddle is VSS.

Pin	Signal	Dir	Note	Pin	Signal	Dir	Note
1	vssa2	power	ground	33	QSPI_CSN0	out	QSPI flash chip select 0 (boot flash)
2	ADC_VINN	analog	SAR ADC negative input, 0 to 1.8 V	34	QSPI_IO0	bidir	QSPI data 0
3	PWM_AH	out	phase A high-side gate	35	QSPI_IO1	bidir	QSPI data 1
4	PWM_AL	out	phase A low-side gate	36	QSPI_IO2	bidir	QSPI data 2
5	PWM_BH	out	phase B high-side gate	37	QSPI_IO3	bidir	QSPI data 3
6	PWM_BL	out	phase B low-side gate	38	vssa1	power	ground
7	PWM_CH	out	phase C high-side gate	39	vssd1	power	ground (user area)
8	PWM_CL	out	phase C low-side gate	40	vdda1	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
9	vdda2	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V	41	QSPI_CSN1	out	QSPI chip select 1 (second device)
10	vssd2	power	ground	42	UART0_TX	out	console UART transmit; boot banner DG32" appears here"
11	PWM_TRIG	out	ADC sample trigger (centre-aligned)	43	UART0_RX	in	console UART receive
12	ENC_A	in	quadrature encoder A	44	UART1_TX	out	UART 1 transmit
13	ENC_B	in	quadrature encoder B	45	UART1_RX	in	UART 1 receive
14	ENC_Z	in	encoder index	46	SPI_SCLK	out	SPI master clock (external ADC / driver)
15	HALL_A	in	Hall sensor A	47	vdda1	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
16	HALL_B	in	Hall sensor B	48	SPI_MOSI	out	SPI master data out
17	vddio	power	3.3 V pad-ring / ESD supply — REQUIRED	49	vccd1	power	1.8 V user-area supply: all DG32-LITE logic, SRAM and the SAR ADC — REQUIRED
18	vccd	power	1.8 V harness core supply — REQUIRED	50	SPI_MISO	in	SPI master data in
19	N/C	power	no connect	51	SPI_CSN	out	SPI master chip select
20	vssa	power	ground	52	vssa1	power	ground
21	resetb	power	harness reset input, active low, 3.3 V — ANDed into the DG32-LITE core reset; may be left pulled up if RST_N (pin 28) is used	53	I2C_SCL	bidir (open-drain)	I2C clock; external pull-up required
22	HALL_C	in	Hall sensor C	54	I2C_SDA	bidir (open-drain)	I2C data; external pull-up required
23	vssd	power	ground	55	GPIO0	bidir	general-purpose I/O 0
24	TCK	in (weak pull-down)	JTAG clock	56	vssio	power	ground
25	TMS	in (weak pull-up)	JTAG mode select	57	GPIO1	bidir	general-purpose I/O 1
26	TDI	in (weak pull-up)	JTAG data in	58	GPIO2	bidir	general-purpose I/O 2
27	TDO	out (driven only while shifting)	JTAG data out	59	FAULT_N	out	lockstep / supervisor fault, active low
28	RST_N	in	external reset, active low (ANDed on die with resetb and the harness POR)	60	RAIL_OK0	in	rail supervisor input 0 (rail_ok[3:2] tied high on die)
29	vssio	power	ground	61	RAIL_OK1	in	rail supervisor input 1
30	vdda	power	3.3 V harness analog supply (POR) — REQUIRED	62	ADC_VINP	analog	SAR ADC positive input, 0 to 1.8 V (analog pad, digital buffer off)
31	CLK	in	system clock, 3.3 V CMOS, 48 MHz nominal (core clock)	63	vccd2	power	1.8 V user supply, unused by DG32-LITE — tie to 1.8 V
32	QSPI_SCLK	out	QSPI boot flash clock	64	vddio	power	3.3 V pad-ring / ESD supply — REQUIRED

3.1 Pins by function

Function	QFN pins
Clock / reset	CLK=31, RST_N=28, resetb=21

Function	QFN pins
QSPI boot flash	SCLK=32, CSN0=33, IO0=34, IO1=35, IO2=36, IO3=37, CSN1=41
UART	UART0 TX=42 RX=43; UART1 TX=44 RX=45
SPI master	SCLK=46, MOSI=48, MISO=50, CSN=51
I2C	SCL=53, SDA=54 (open-drain, external pull-ups)
GPIO	GPIO0=55, GPIO1=57, GPIO2=58
Supervisor	FAULT_N=59, RAIL_OK0=60, RAIL_OK1=61
SAR ADC (analog)	VINP=62, VINN=2 (0 to 1.8 V)
Motor PWM	AH=3, AL=4, BH=5, BL=6, CH=7, CL=8, TRIG=11
Position sensing	ENC_A=12, ENC_B=13, ENC_Z=14, HALL_A=15, HALL_B=16, HALL_C=22
JTAG	TCK=24, TMS=25, TDI=26, TDO=27
Attention (internal)	memory-mapped at 0xE000_0000; clk_fast_i is a review item

4 Electrical characteristics

Pre-silicon. Recommended operating values are the sky130 process nominals for this die's supply domains; absolute-maximum values are conservative process limits, not device characterisation. First-silicon characterisation will replace this section.

4.1 Recommended operating conditions

Parameter	Sym	Min	Typ	Max	Unit	Note
Core supply	vccd1	1.71	1.8	1.89	V	sky130 core, +/-5%; powers all logic, SRAM and the SAR ADC
I/O supply	vddio	3.0	3.3	3.6	V	3.3 V CMOS pad ring / ESD
Core clock	CLK	-	50	~55	MHz	fmax ~55-62 MHz measured post-route; 50 MHz design target
ADC input	VINP/VIN N	0	-	1.8	V	differential, vccd1 domain
Junction temp	TJ	-	25	-	degC	power quoted at tt 25 C; range TBD on characterisation

4.2 Absolute maximum ratings (process limits, pre-characterisation)

Parameter	Min	Max	Unit
Core supply vccd1	-0.3	1.95	V
I/O supply vddio	-0.3	3.63	V
Voltage on any digital pad	-0.3	vddio + 0.3	V
Analog pad voltage	-0.3	vccd1 + 0.3	V
Storage temperature	-40	125	degC (target)

4.3 Power

Estimated total ~0.43 W at 50 MHz (OpenROAD vectorless, tt 25 C, 1.8 V). All dynamic and leakage current is drawn from vccd1; the other user supply domains are unused by DG32-2DOM and are tied only to keep the pad ring's domain-order ESD structures biased.

5 Power supplies and sequencing

Rail	QFN pins	Voltage	Purpose
vddio	17, 64	3.3 V	pad-ring / ESD supply - REQUIRED
vccd	18	1.8 V	harness core supply - REQUIRED
vccd1	49	1.8 V	user-area supply: all logic, SRAM, SAR ADC - REQUIRED
vdda	30	3.3 V	harness analog / POR supply - REQUIRED
vdda1	40, 47	3.3 V	user analog, unused - tie to 3.3 V
vdda2	9	3.3 V	user analog, unused - tie to 3.3 V
vccd2	63	1.8 V	user digital, unused - tie to 1.8 V
vss* / paddle	1,10,20,23,29,39,52,56 ...	0 V	all grounds and the exposed paddle

Sequencing. Bring up 3.3 V (pad ring) before or together with 1.8 V, as on the ChipFoundry development board. All vss* pins and the exposed paddle go to ground. DG32-2DOM draws its whole current from vccd1 (1.8 V); tie the unused user rails (vdda1, vdda2, vccd2) to their nominal voltage so the ESD structures stay biased.

6 Clock, reset and boot

6.1 Clock and reset

A single core clock enters on CLK (pin 31), 3.3 V CMOS, 50 MHz design target. There is no on-die PLL. Two reset inputs are ANDed on die: the OpenFrame harness resetb (pin 21, active low) and RST_N (pin 28, active low); either may be used, pull the unused one high. DG32-2DOM adds a second clock, clk_fast_i (114 MHz), for the attention engine; the CPU and all peripherals stay in the 50 MHz domain and the two are bridged by CDC logic.

6.2 Boot flow

The die is standalone - no management core. At power-on the on-die 64 KB boot ROM runs with no external help:

- Emit the boot banner on UART0 (pin 42) at 115200 8N1 (baud divider fixed for 50 MHz).
- Read the QSPI NOR flash header through the 0x1 window (boot magic 0xD632_B007, then length).
- Copy the flash image into SRAM over the QSPI quad-output-fast-read (0x6B, 1-1-4, 8 dummy cycles, SCLK = clk/2).
- Jump to the copied image and run from SRAM/ROM.
- A UART monitor (r/w/j/i commands) is available if the flash is blank, so a bare board is still inspectable.

Instruction fetch is ROM-only. The CPU fetch bus is wired to the boot ROM alone. The QSPI-XIP window and SRAM are reachable by loads and stores, not by instruction fetch - code does not execute in place from external NOR, it is copied to SRAM first. Firmware and linker scripts must place executable code in SRAM.

7 Memory map

Base-address decode from the top-level decoder: selector is addr[31:28]; the 0xD page sub-decodes addr[27:24], so each 0xD peripheral base is 0xDX00_0000. Any unmapped access completes with SLVERR (it never hangs the bus).

Base	Block	Note
0x1000_0000	QSPI-XIP read window	data read window (loads); NOT instruction fetch
0x9000_0000	SRAM	32 KB main data memory
0xC000_0000	DShot ESC engine	pin-muxed onto the PWM pads
0xD000_0000	UART0	console / boot banner
0xD100_0000	UART1	telemetry
0xD200_0000	SPI master	
0xD300_0000	GPIO	14 pins on the die
0xD400_0000	TIMER0	
0xD500_0000	TIMER1	
0xD600_0000	Fault CSR	poison + cause
0xD700_0000	Supervisor / wWDt	
0xD800_0000	PWM	3-phase center-aligned
0xD900_0000	I2C master	
0xDA00_0000	Encoder	quadrature + Hall
0xDB00_0000	SAR ADC	8-bit differential
0xDC00_0000	DMA	
0xDD00_0000	SYSCTL	device ID / clock gate
0xDE00_0000	CORDIC	Q1.31
0xDF00_0000	IRQ aggregator	
0xE000_0000	INT8 attention	DG32-2DOM only (INCLUDE_ATTN)

Address constraint (firmware). The CPU has a hardwired PMA whitelist: a load or store is legal only if addr[31] is set or the address is in 0x1xxx_xxxx. That is why SRAM is at 0x9 and the peripherals at 0xC/0xD/0xE. An access outside the whitelist faults inside the CPU before the bus sees it.

8 Register map

Byte offsets within each block, extracted from the RTL and cross-checked against the AXI decode. Sticky bits clear as noted (W1C = write 1 to clear). Self-clearing start bits are marked in the block.

UART0 / UART1 (0xD0 / 0xD1)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [31:16]=clkdiv (baud = clk_i/clkdiv)
0x04	TX	W	[7:0]=byte to send (8N1); write back-pressures while tx_busy - never drops a byte
0x08	RX	R	[7:0]=received byte; read clears rx_valid
0x0C	STATUS	R	[0]=tx_busy; [1]=rx_valid

8N1: 1 start + 8 data LSB-first + 1 stop. RX 2-flop synchronised. 115200 baud at 50 MHz = 434 clocks/bit.

SPI master (0xD2)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=cpol; [2]=cpha; [9:4]=xfer_bits (8/16/32); [31:16]=clkdiv
0x04	TX	W/R	write launches a full-duplex MSB-first transfer when enabled and not busy
0x08	RX	R	last received word
0x0C	STATUS	R	[0]=busy
0x10	CS	R/W	[0]=cs_n level (reset 1 = deasserted)

SCLK idle level = CPOL. Transfer = 2 x xfer_bits SCLK edges. Hardware master, 25 MHz, 0.64 us per 16-bit word.

GPIO (0xD3)

Offset	Name	R/W	Bits / meaning
0x00	DATA_OUT	R/W	[N-1:0] value driven on pins whose OE bit is set
0x04	OE	R/W	[N-1:0] 1=drive, 0=high-Z input (reset 0 = all inputs); active-high
0x08	DATA_IN	R	[N-1:0] 2-flop synchronised pin state; writes accepted and discarded
0x0C	SET	W1S/R	write 1 sets that DATA_OUT bit; reads as DATA_OUT
0x10	CLR	W1C/R	write 1 clears that DATA_OUT bit; reads as DATA_OUT

SET/CLR are separate addresses onto one DATA_OUT register (atomic single-pin toggling). N=14 on the die.

TIMER0 / TIMER1 (0xD4 / 0xD5)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=auto_reload; [2]=irq_enable; write [8]=clear pending (W1C)
0x04	RELOAD	R/W	period-1 (fires every RELOAD+1 clocks)
0x08	COUNT	R	current down-counter value
0x0C	STATUS	R	[0]=pending (cleared via CTRL[8])

Down-counter; pulses at 0; irq = pending and irq_enable. Enable-rising reloads count.

Fault CSR (0xD6)

Offset	Name	R/W	Bits / meaning
0x0	STATUS	R	[0]=poison; [3:1]=cause. Sticky; cleared only by reset
0x4	INJECT	W	write MAGIC 0xDEAD_5AFE -> corrupt pulses 32 cycles. Reads 0
0x8	EVENTS	R	matched lockstep-compare count
0xC	ID	R	0xD632_0001

cause 001 = DATA_MISMATCH. Sticky-first: holds the first cause across continued execution.

Supervisor / wWDT (0xD7)

Offset	Name	R/W	Bits / meaning
0x00	ID	R	0x00D6_0006
0x04	CTRL	R/W	[0]=global_en; [1]=wdt_en; [7:4]=rail_en[3:0]
0x08	DEGLITCH	R/W	[7:0] deglitch cycles (reset 8)
0x0C	HOLDOFF	R/W	[15:0] reset-release holdoff (reset 1000)
0x10	WDTMIN	R/W	[15:0] watchdog window min (reset 100)
0x14	WDTMAX	R/W	[15:0] watchdog window max (reset 50000)
0x18	KICK	W	write 0xA5A5 = valid watchdog kick
0x1C	STATUS	R/W1C	[0]=sys_rst_n; [1]=wdt_fault; [7:4]=rail_sticky; [15:8]=fault_cnt

Windowed watchdog: a kick before WDTMIN or after WDTMAX both fault. Arms only when firmware sets global_en.

PWM (3-phase) (0xD8)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=EN
0x04	PERIOD	R/W	counter top, 0..65535 (period = 2*(PERIOD+1) cycles, center-aligned)
0x08	DEADTIME	R/W	dead-time cycles, all 3 legs (8-bit, max 255 = 5.1 us at 50 MHz)
0x0C	DUTY_A	R/W	phase A shadow duty, 0..PERIOD
0x10	DUTY_B	R/W	phase B shadow duty
0x14	DUTY_C	R/W	phase C shadow duty
0x18	STATUS	R	[0]=BRAKED (sticky)
0x1C	BRAKE_CLR	W	write [0]=1 to clear BRAKED once brake_i is low again

Shadow duties latch at cnt==0; sample_trig pulses there. Async hardware brake forces all 6 outputs off in <=2 cycles.

I2C master (0xD9)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [31:16]=clkdiv (SCL = clk_i/(4*(clkdiv+1))); reset 124 -> 100 kHz
0x04	CMD	W/R	[7:1]=7-bit addr; [0]=read/write; [15:8]=byte count 1..4; [16]=send STOP
0x08	TXD	R/W	up to 4 bytes to send, LSB first on the wire
0x0C	RXD	R	up to 4 bytes received, LSB first
0x10	STATUS	R/W1C	[0]=busy; [1]=nack_addr; [2]=nack_data; [3]=arb_lost (write 1 to clear)

7-bit, 100/400 kHz, clock stretching, open-drain (external pull-ups). 2-byte read ~72 us at 400 kHz.

Encoder (quad + Hall) (0xDA)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=EN (gates quad decode + Hall capture)
0x04	POSITION	R/W	signed 4x quadrature position; firmware may write to preset/zero
0x08	INDEX_POSITION	R	POSITION latched at last Z rising edge
0x0C	STATUS	R/W1C	[0]=INDEX_SEEN (sticky); [1]=DIR (live)
0x10	TIMESTAMP	R	free-running 32-bit cycle counter
0x14	HALL_STATE	R	[2:0]=synchronised {hall_c, hall_b, hall_a}
0x18	HALL_TIMESTAMP	R	TIMESTAMP at last Hall transition
0x1C	HALL_EVENT	R/W1C	[0]=NEW_EVENT (sticky)

Edge timestamps give velocity from time-between-edges, accurate at low speed.

SAR ADC (0xDB)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=cal; [31:16]=clkdiv (reset 24 -> 50 MHz/25 = 2 MHz)
0x04	START	W	any write starts one conversion when enabled and not busy
0x08	RESULT	R	[7:0]=last conversion; [8]=valid (cleared on START)
0x0C	STATUS	R	[0]=busy; [1]=valid; [2]=comp (raw comparator)
0x10	TRIM	R/W	[4:0]=trim bits; trimb driven as complement (reset 01111)

8-bit differential SAR hard macro (1.8 V). ~200 kSa/s at reset divider; per-die trim is a bring-up step.

DShot ESC engine (0xC000_0000)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	W	[0]=START (self-clearing; ignored while BUSY)
0x04	BITPERIOD	R/W	cycles per bit slot (12-bit)
0x08	T0H	R/W	high-time for logical 0
0x0C	T1H	R/W	high-time for logical 1
0x10	THROTTLE0	R/W	[10:0]=throttle; [11]=telemetry request; channel 0
0x14	THROTTLE1	R/W	channel 1
0x18	THROTTLE2	R/W	channel 2
0x1C	THROTTLE3	R/W	channel 3
0x20	STATUS	R	[0]=BUSY

16-bit frame MSB-first: [15:5] throttle, [4] telemetry, [3:0] CRC4. All 4 channels share one bit-time counter.

CORDIC (0xDE)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=START (self-clearing); [1]=MODE (0=rotation, 1=vectoring)
0x04	XIN	R/W	Q1.31 x input
0x08	YIN	R/W	Q1.31 y input
0x0C	ZIN	R/W	Q1.31 angle/pi
0x10	XOUT	R	Q1.31 result x (rotation: cos; vectoring: magnitude/K)
0x14	YOUT	R	Q1.31 result y (rotation: sin)
0x18	ZOUT	R	Q1.31 result z (vectoring: atan2/pi)
0x1C	STATUS	R	[0]=BUSY

N=20 iterations, ~53-58 cycles/op. K=0x4DBA76D4; vectoring magnitude is not gain-compensated by design.

IRQ aggregator (0xDF)

Offset	Name	R/W	Bits / meaning
0x00	PENDING	R/W1C	[15:0] pending; write 1 to clear
0x04	ENABLE	R/W	[15:0] mask; external int = OR(PENDING and ENABLE)
0x08	MSIP	R/W	[0] -> machine software interrupt
0x0C	RAW	R	[15:0] source levels before edge detection
0x10	SWSET	W	write 1 sets PENDING[7]
0x14	ID	R	0x00D6_1249

Sources: 0 TMR0, 1 TMR1, 2 PWM trig, 3 ADC done, 4 DMA done, 5 I2C done, 6 FAULT, 7 SW, 8 ATTN (2DOM).

DMA (0xDC)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	R/W	[0]=enable; [1]=start (self-clearing); [2]=src_incr; [3]=dst_incr
0x04	SRC	R/W	source address, word-aligned
0x08	DST	R/W	destination address, word-aligned
0x0C	LEN	R/W	transfer length in words (0 = none)
0x10	STATUS	R/W1C	[0]=busy; [1]=done (write 1 to clear); [2]=error
0x14	COUNT	R	words remaining

Single-beat AXI master; the top arbitrates LSU-priority over DMA. ~7.1 cycles/word.

INT8 attention engine (0xE000_0000) - programming (writes)

Offset	Name	R/W	Bits / meaning
0x00	CTRL	W	[0]=start (self-clearing); [1]=SKIP_LOAD (reuse EXP/K/V from previous kick)
0x04	A_BASE	W	Q base byte address
0x08	A_PITCH	W	Q position pitch (bytes)
0x0C	B_BASE	W	K base byte address
0x10	B_PITCH	W	K position pitch (bytes)
0x14	D_BASE	W	O (output) base byte address
0x18	D_PITCH	W	O position pitch (bytes)
0x1C	M	W	MROWS: query rows this kick (1..65535)
0x20	N	W	NK: keys (1..MAXK=400)
0x24	K	W	KD: Q/K head dim in bytes (mult of LANES and 4)
0x28	SHIFT	W	EXP index right-shift (6-bit)
0x2C	C_BASE	W	V base byte address
0x30	C_PITCH	W	V position pitch (bytes)
0x34	DV	W	output channels/head in bytes (mult of LANES and 4)
0x38	E_BASE	W	EXP table base (256 u15 entries)
0x3C	RATIO	W	u32 16.16 window constant; rq = floor((RATIO<<16)/Z)

INT8 attention engine (0xE000_0000) - status (reads)

Offset	Name	R/W	Bits / meaning
0x00	STATUS	R	[0]=busy; [1]=done; [2]=Z==0 seen; [3]=rq overflow; [4]=EXP clamp; [5]=geometry rejected
0x40	Z	R	Z of the last completed row
0x44	rq	R	rq of the last completed row

Programming a kick (firmware). Write the Q/K/V/O base and pitch, the geometry (M rows, N keys, KD, DV), SHIFT, the EXP table base and RATIO, then set CTRL[0]=start. Poll STATUS[0]=busy to 0; STATUS[1]=done. SKIP_LOAD (CTRL[1]) reuses the K/V/EXP buffers from the previous kick. The engine runs on clk_fast_i (114 MHz) and reaches memory through the CDC bridges, so it does not stall the 50 MHz CPU.

9 Board design guidance (hardware / PCB)

- Motor half-bridge gate driver on PWM_AH..CL (pins 3-8); the outputs are complementary with programmable dead-time. Respect the driver's shoot-through spec; the die also has a hardware brake that forces all six low.
- Route PWM_TRIG (pin 11) to nothing external unless you want the ADC-sample instant on a test point; it fires at the centre-aligned current-ripple null.
- Wire the encoder A/B/Z (pins 12-14) and Hall A/B/C (pins 15,16,22) directly; inputs are 2-flop synchronised and 3.3 V.
- SAR ADC inputs ADC_VINP (62) and ADC_VINN (2) are differential, 0 to 1.8 V - scale phase-current shunt amplifiers into that window and keep the pair matched and short.
- QSPI NOR boot flash on pins 32-37 (add CSN1=41 for a second device / PSRAM); a 3.3 V W25Q / S25FL works. Without a flash the ROM drops to the UART monitor.
- I2C_SCL (53) and I2C_SDA (54) are open-drain - the carrier must supply the pull-ups (typ 2.2-4.7 k to 3.3 V).
- Bring UART0 (42/43) to a USB-UART for the boot banner and console; UART1 (44/45) is free for telemetry.
- FAULT_N (59) is an active-low fault output from the lockstep/supervisor - route it to the driver enable or a supervisor for a hardware trip.
- Provide 3.3 V and 1.8 V per Section 5; the 100 production parts need a carrier with the 64-pin QFN footprint (KiCad from chipfoundry/caravel_board), gate drives, encoder, an optional external SPI ADC, and UART/JTAG headers.
- The attention engine adds no board signals; budget for clk_fast_i only if the CI2612 die brings it out to a pad rather than deriving it on-die.

JTAG idles safely with no probe. TMS (pin 25) and TDI (pin 26) are weakly pulled up on die, TCK (pin 24) weakly pulled down, so an unconnected JTAG header will not disturb the part.

10 Package

Item	Value
Package	64-pin QFN
Body	9.0 x 9.0 mm
Pitch	0.5 mm
Exposed paddle	VSS (solder to ground plane)
Pin 1	upper-left, counter-clockwise numbering
Process	SkyWater sky130A, 130 nm
Shuttle	chipIgnite CI2612

Preliminary datasheet, pre-silicon. Register and pin detail from the taped-out RTL (CI2609 R2) and the QFN-64 pin map derived from the OpenFrame harness pinout, awaiting ChipFoundry bond-diagram confirmation. Electrical limits are design/process nominals pending first-silicon characterisation. DG32-2DOM adds the INT8 attention engine and a second (114 MHz) clock domain; its A3 die closed with core +0.30 ns at 50 MHz and clk_fast +0.19 ns at 114 MHz.